

AI Executive Assistant Agent

Email management and meeting scheduling

Sprint 3 goal: Generation of draft responses and integration of supabase and Langchain



Sprint Goal Recap

Objective

- Generate draft response
- integrate LangChain
- Integrate Supabase



Technical Decisions (1)

Draft response

- Drafts are saved via the Gmail API (`drafts.create`) and are never sent automatically — human control is maintained
- Three distinct prompt branches depending on the context: free calendar slot → confirmation, occupied slot → suggestion of alternatives, missing date/time → request for information
- The LLM model generates the text; `gmail_client.save_draft()` pushes it into Gmail
- Language detection in the classifier: the 'language' field is extracted once the entire body has been read, and then inserted as a hard constraint into the draft's system prompt

Supabase integration

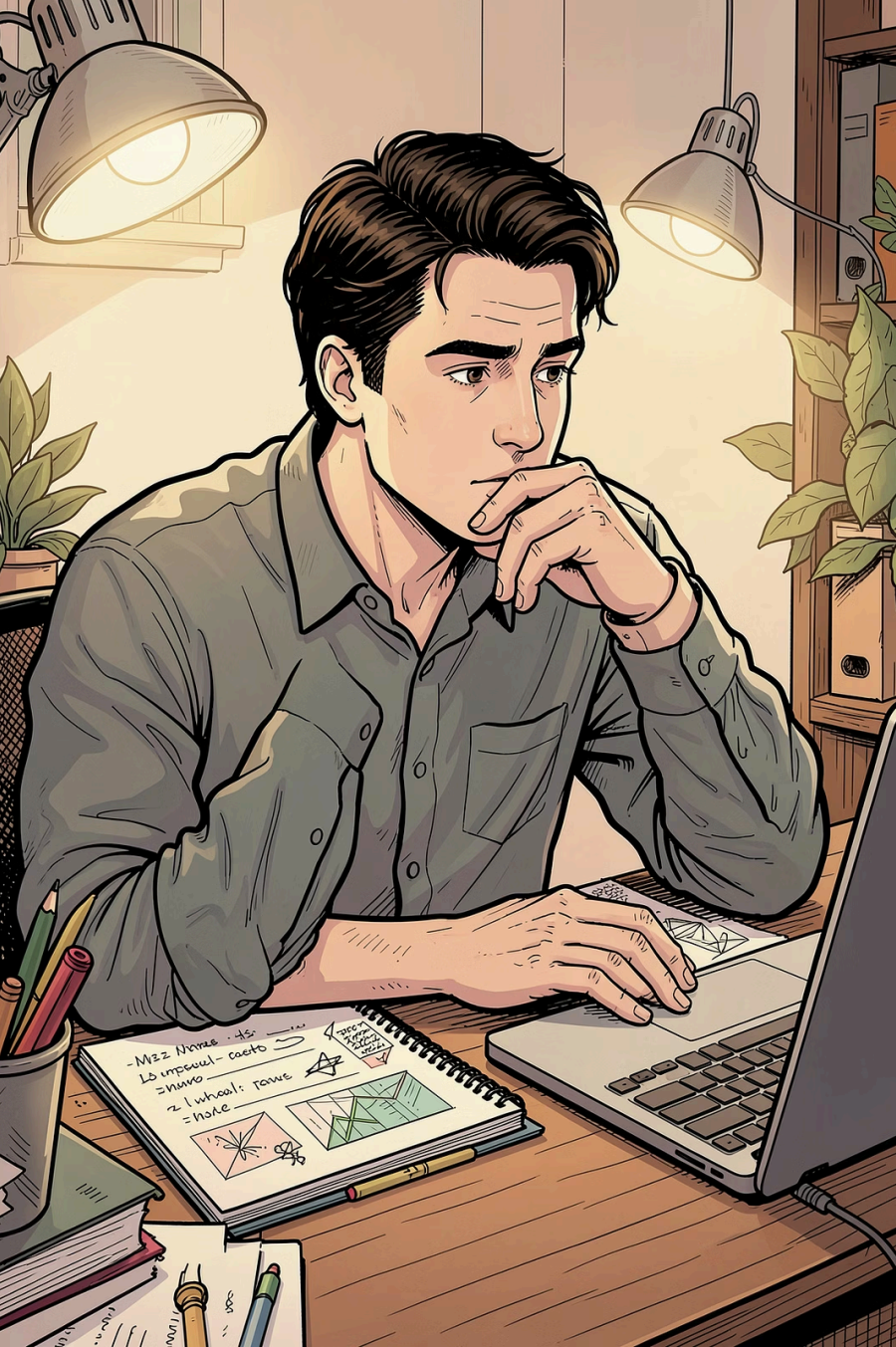
- `email_memory` table with the following fields: `message_id`, `subject`, `sender`, `intent`, `status`
- `save_email(...)` after processing status encodes the result: DRAFTED, IGNORED, or ERROR

Technical decisions (2)

LangChain integration

- create_agent + tools — orchestration via the LLM Decision: replace the deterministic Python pipeline with a LangChain agent that determines the order of tool calls at runtime.
- The tools wrapped existing modules: fetch_unread_emails, classify_email, check_and_book_calendar, generate_and_save_draft, check_if_processed, mark_as_processed, mark_email_as_read Challenges

Tool	Action
is_processed	check if the email has already been processed
classify_email	Intent + date + time + language
book_calendar	Check availability + create event
generate_draft	Draft reply
save_draft	Save to Gmail
mark_as_read	Gmail label
save_to_memory	Save Supabase result



Learnings

Biggest challenges

- Token budget: with the deterministic pipeline = 2 LLM calls per email. The ReAct agent need more Tokens per email (think → tool → observe loop).
- Integration of LangChain: `create_react_agent` is deprecated → `create_agent`
- Response language

Next Sprint Plan

Planned features for the next sprint:



Memory

Manage preferences and vip sender



Implementation of UI

Implement an UI that allows users to view the draft reply and to approve or reject the sending of the email



Live demo

Key moments of the demo

- Draft response
- Supabase table
- langchain tools

A stylized illustration of a woman with long brown hair, wearing a light-colored blazer over a white shirt. She is pointing her right index finger towards the left. Above her head are two thought bubbles, each containing a question mark. To her right is a wooden signpost with a horizontal arrow pointing right, also containing a question mark. The background features a network of thin lines connecting various nodes, suggesting a digital or conceptual space. The overall color palette is warm, with shades of beige, cream, and light brown.

Open Questions / Feedback Needed

- UI better than response in the draft?